

SKANDA GONUR NAGARAJ

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EDUCATION

San Jose State University <i>Master of Science in Computer Science</i> Coursework: Distributed Systems, Cloud Computing, Database Systems, NoSQL, Machine Learning, Artificial Intelligence	Aug 2024 – May 2026 GPA: 3.5/4.0
Visvesvaraya Technological University <i>Bachelor of Engineering in Computer Science</i> Coursework: Analysis and Design of Algorithms, Data Structures, Database Systems, Operating Systems, Computer Networks, Advanced Java, Software Engineering, Computer Organization, OOPs	Aug 2019 – Jun 2023 GPA: 3.97/4.0

WORK EXPERIENCE

Research Assistant <i>San Jose State University</i> - Building a Legal Document Analysis system processing 9,280+ civil rights cases; designed a modular NLP pipeline covering data ingestion, preprocessing, model inference, and structured output generation, and engineered a long-context inference pipeline handling documents exceeding 75,000 tokens via chunking strategies, context windowing, and input routing across BART, LED, and LongT5 architectures. - Built a benchmarking and evaluation framework running 11 model configurations with automated metric collection and structured experiment logging; evaluated domain-specific vs. general-purpose NLP models enabling reproducible comparisons across architectures, hyperparameter settings, and overfitting behavior.	Aug 2025 – May 2026
Software Engineer <i>Vodafone Intelligent Solutions (VOIS)</i> - Developed Spring Boot microservices and REST APIs (Controller/Service/Repository) for core workflows deployed on AWS ECS with Docker, contributing to systems handling 5M+ requests/day with p95 <300ms latency. - Optimized MongoDB queries on collections exceeding 50M documents by adding compound indexes and refactoring aggregation pipelines, reducing API response latency by 25% under peak traffic. - Integrated a third-party payment gateway with idempotent request handling, exponential backoff retries, and circuit-breaker patterns via Resilience4j to prevent duplicate charges and handle downstream failures gracefully. - Worked on Kafka-based async event streaming for core workflows, helping decouple services and implementing dead-letter topic handling to ensure zero message loss during consumer downtime. - Contributed to GitLab CI/CD pipelines for microservices deployed on AWS ECS, helping reduce average deployment time from 45 to 12 minutes and cut failed deployments by 40% through automated integration tests and staged rollouts. - Secured 40+ REST endpoints with Spring Security + JWT and documented APIs using Swagger/OpenAPI, reducing integration defects by 30% and supporting faster onboarding for downstream teams.	Aug 2023 – Aug 2024
Machine Learning Intern <i>Ekathva Innovations Pvt. Ltd.</i> Built an end-to-end customer risk classification pipeline on 3M+ transaction records — covering feature engineering, model training, and threshold tuning — improving recall from 62% to 78% while maintaining 75% precision.	Aug 2022 – Oct 2022

SKILLS

Languages: Java, Python, C++ , TypeScript, Swift Backend: Spring Boot, Spring MVC, Spring Security, REST APIs, JWT, Swagger/OpenAPI, FastAPI Cloud & DevOps: AWS (EC2, Lambda, S3, API Gateway, Cognito), Docker, Kubernetes, Terraform, GitLab CI/CD, Maven Data & Streaming: Kafka, PySpark, Hadoop, Apache Mahout, NumPy, Pandas Databases & Tools: MongoDB, PostgreSQL, MySQL, Redis, Elasticsearch, Kibana, Postman, JIRA Additional: LangChain, RAG, Hugging Face Transformers, PyTorch
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PROJECT EXPERIENCE

Cadence: Live-Coding DSL for Real-Time Music Generation - Designed and implemented Cadence, a custom domain-specific language (DSL) for live music coding with a full execution pipeline — lexer, recursive-descent parser, AST, and tree-walking interpreter — capable of parsing and executing programs in under 5ms. - Built a multi-track concurrent runtime using Python threading with beat-accurate timing ($\pm 2\text{ms}$ jitter), NumPy/Pygame audio synthesis, and hot-swap file watching that reflects code changes to live audio in under 100ms without restarting playback. - Validated interpreter correctness with a unit test suite covering 95%+ of grammar rules; benchmarked runtime under concurrent track load, sustaining 8 simultaneous tracks with no audio dropout at 120–180 BPM.	April 2026
Distributed LLM-Powered Document Retrieval System - Designed and deployed a distributed document retrieval platform using FastAPI microservices with a modular architecture supporting parallel embedding generation and concurrent query execution. - Built a high-throughput backend sustaining 300+ QPS (peak 900) with p95 latency <650 ms; implemented vector storage in Redis and optimized query routing for low-latency retrieval. - Containerized microservices with Docker and orchestrated deployment on AWS EC2, achieving 99.95% uptime and end-to-end latency <500 ms (p95) across distributed nodes.	Jan 2025 – Feb 2025
AudioTranscriber: Real-Time Speech-to-Text iOS App - Built a modular SwiftUI iOS app using AVAudioEngine; designed a parallel audio processing pipeline with 30-second segmentation achieving <2s end-to-end transcription latency. - Engineered a dual-backend failover system (OpenAI Whisper API + on-device SFSpeechRecognizer) reducing failed transcriptions by 40% and enabling 100% offline feature support. - Designed a SwiftData persistence layer with 3 relational models; optimized via background writes, lazy loading, and file-path storage, reducing memory usage by 60% and supporting sessions up to 4+ hours.	Sept 2025
TastyThreads: Location-Based Discussion Platform - Built a full-stack location-based platform (threads + posts) and deployed on AWS using Terraform (IaC) and CI/CD, enabling consistent, repeatable releases with automated builds and deployments. - Implemented secure authentication with Amazon Cognito and a serverless backend (Lambda + API Gateway) with S3 media storage; added health checks and error handling to improve reliability and reduce manual ops.	Mar 2025 – May 2025

- Optimized geospatial search by integrating **Google Maps + OpenStreetMap** APIs and adding caching/request batching, reducing search response time by **45%**.
- Scalable Email Content Analysis with MapReduce + Clustering

Dec 2024

 - Built a **Hadoop MapReduce** pipeline to parse and structure **500K+ unstructured email messages** into analytics-ready datasets, reducing ingestion time by **35%** via optimized parsing and combiner-side aggregation.
 - Applied **Apache Mahout K-Means** to cluster emails using TF-IDF/keyword features, improving segmentation quality by **20%** (purity/silhouette) over baseline configuration.
 - Improved preprocessing robustness by cleaning noisy tokens and filtering outliers, increasing clustering precision and consistency by **15%** across evaluation runs.

- Customer Churn Analysis in Telecom Sector

Oct 2024 – Nov 2024

 - Designed an end-to-end ML pipeline for churn prediction covering preprocessing, feature engineering, and EDA on **7K+ customer records**, cutting manual data cleaning time by **40%**.
 - Trained and fine-tuned Logistic Regression, Random Forest, and **Deep Learning (TensorFlow)** models, achieving a **93% F1-score** and **28% improvement in recall** compared to the baseline model.

PUBLICATIONS

Machine-learning and Deep-learning Based Parkinson Disease Detection System – IJSREM, ISSN: 2582-3930, June 2023.

A Review on Detection of Parkinson’s Disease Using ML Algorithms – IJRASET, ISSN: 2321-9653, March 2023.

ACHIEVEMENTS

Finalist — Scoop AI Hackathon (Silicon Valley Bowl), hosted by **UC Berkeley & Stanford**